

Activity-4

V Deepa Dharshini
(192525213)

1) Given, {4371, 1323, 6173, 4199, 4344, 9679, 1989}.

hash function : $h(x) = x \% 10$

Hash values: 4371 $\rightarrow$ 1

1323 $\rightarrow$ 3

6173 $\rightarrow$ 3

4199 $\rightarrow$ 9

4344 $\rightarrow$ 4

9679 $\rightarrow$ 9

1989 $\rightarrow$ 9

a) Separate Chaining

Index	values.
0	-
1	4371
2	-
3	1323 $\rightarrow$ 6173
4	4344
5	-
6	-
7	-
8	4199 $\rightarrow$ 9679 $\rightarrow$ 1989
9	-

(b) Linear Probing:

Index	Value
0	—
1	4371
2	—
3	1323
4	6173
5	4344
6	9679
7	1939
8	—
9	4199

(c) Quadratic Probing

Index	Value
0	—
1	4371
2	—
3	1323
4	6173
5	4344
6	9679
7	1939
8	—
9	4199

(d) Double Hashing

Index	Value
0	—
1	4371
2	6173
3	1323
4	4344
5	9679
6	—
7	—
8	1989
9	4199

19/8/20